

Flight-Data Replay Bench for Actuator Endurance Testing

Turn a real flight into a repeatable endurance test, and know exactly how many hours every servo has flown.

- 1 Problem statement.** Build a web-based controller that converts recorded drone flight logs into servo command profiles, replays them over Wi-Fi on a 16-channel PWM test bench, and tracks cycles completed and per-channel running hours so engineers can quantify actuator life. A bench simulator, protocol spec and sample logs are provided; teams demo on real hardware.

BACKGROUND

Drone servos, ESCs and linkages wear out, and the only way to learn how long a part lasts is to fly it: slow, expensive, unrepeatable. Welkinrim is building a bench that plays a recorded flight back into the real actuators, over and over, and counts every hour and every cycle. The hardware exists: an ESP32-S3 drives a PCA9685 16-channel PWM board, shows status on an OLED and exposes a WebSocket API over Wi-Fi. What is missing is the software an engineer sits in front of. That is your 24 hours.

WHAT WE PROVIDE

- **Bench simulator** (Python, one file) speaking the exact protocol of the real bench, including link drops and power-cycle behaviour.
- **Protocol specification:** messages, profile file format, error cases.
- **Sample flight logs:** CSV exports and raw ArduPilot `.bin` files.
- **Three real benches at the venue** (ESP32-S3 + PCA9685 + servos + OLED) on a shared Wi-Fi network for testing, demos and judging.

RULES

Any language or framework; the GUI must run in a browser. Firmware is fixed and not part of this competition (see Competition B). Work against the simulator all night, book bench slots to test on hardware. **Grand finale:** the winning controller drives the winning firmware from Competition B live on stage.

WHAT YOU BUILD: A PC CONTROLLER WITH A BROWSER GUI

1. **Import:** load a flight log, pick 1–16 channels, resample to the bench frame rate (50–400 Hz), preview traces, push the profile to the bench.
2. **Control:** Start, Pause, Stop, Reset counters, set a cycle target, set channel pulse widths manually while stopped.
3. **Live dashboard:** link state, run state, cycle N of target, percent of current cycle, live pulse width per channel.
4. **Hour meters:** per-channel bench hours and active hours, synced to wall-clock time through the protocol's time-sync message, never lost on reconnect.
5. **History:** sessions and events (start, stop, cycle complete, link lost/regained) stored in a database, browsable by date range.

Stretch: ArduPilot/PX4 importers, link-loss alerts, several benches on one dashboard, exportable test report.

JUDGING · 100 POINTS

Functional completeness against the must-have list	40
Correctness & robustness: reconnects, counters survive a bench power cycle, time sync	25
Usability of the GUI on the real bench	20
Code quality, README, ease of running	15

Prize. The winning team is offered an internship at Welkinrim, working on the production version of this bench.

